

FinRL-DeepSeek: Enhancing Reinforcement Learning for Stock Trading with Large Language Models

Abstract: This paper presents an advanced automated stock trading agent that integrates DeepSeek, a state-of-the-art Large Language Model (LLM), with Reinforcement Learning (RL). By leveraging the Financial News and Stock Price Integration Dataset (FNSPID), we extract nuanced market sentiment and trading signals using DeepSeek. These signals are then injected into a Proximal Policy Optimization (PPO) agent. Our methodology demonstrates the potential of combining NLP-driven market insights with quantitative RL strategies to navigate complex financial markets.

1. Introduction

Automated stock trading has evolved significantly with the advent of deep reinforcement learning (DRL). However, traditional DRL agents often rely solely on historical price data, ignoring the wealth of information present in financial news. The FinRL Contest 2025 challenges participants to bridge this gap. This paper introduces a novel architecture based on the FinRL-DeepSeek framework, utilizing the DeepSeek model to process financial news and enhance RL trading strategies.

2. Methodology

2.1. Dataset

We utilize the FNSPID (Financial News and Stock Price Integration Dataset), which contains synchronized stock prices and 15 million financial news articles for Nasdaq companies (1999-2023). This provides a robust foundation for both quantitative and qualitative analysis.

2.2. Architecture

The system is composed of two main modules:

- LLM Signal Extraction:** DeepSeek is prompted to analyze daily news articles from the FNSPID dataset. It outputs a sentiment score (-1 to 1) and a risk assessment signal.
- RL Trading Agent:** A PPO (Proximal Policy Optimization) agent is trained on an environment where the state space includes both historical OHLCV (Open, High, Low, Close, Volume) data and the engineered signals from DeepSeek.

2.3. Prompt Engineering for DeepSeek

To extract actionable insights, we designed specific prompts for DeepSeek:

"Analyze the following financial news article for [Company Name].
Provide a sentiment score between -1 (highly negative) and 1 (highly positive).
Assess the short-term risk level (Low, Medium, High)."

News text: {news_content}"

3. Experimental Setup and Expected Results

The model is trained on data from 2010 to 2020 and validated on 2021-2023 data. The state space is defined as $S = [\text{Price_Data}, \text{Technical_Indicators}, \text{DeepSeek_Sentiment}, \text{DeepSeek_Risk}]$. The action space is continuous, representing the portfolio weights. We expect the FinRL-DeepSeek agent to outperform baseline models (e.g., standard PPO without news, Buy-and-Hold strategy) in terms of Sharpe Ratio and Cumulative Return, particularly during periods of high market volatility where news sentiment plays a critical role.

4. Conclusion

Integrating DeepSeek's advanced natural language processing capabilities with reinforcement learning offers a promising avenue for automated trading. By capturing complex market sentiments from the FNSPID dataset, our proposed FinRL-DeepSeek agent aims to achieve superior risk-adjusted returns. Future work will explore more computationally intensive methods, such as adapting DeepSeek-R1 training variants specifically for stock trading.

5. References

- [1] FinRL Contest 2025. Open Finance Lab.
- [2] Zihan1004. Financial News and Stock Price Integration Dataset (FNSPID).
- [3] DeepSeek AI. DeepSeek LLM Documentation.